

KAGGLE COMPETITION: HOUSE PRICES - ADVANCED REGRESSION TECHNIQUES

Group Members:
Varsha Peddapapireddy gari
Ronaldo Cunha
Srija Dasgupta
Kaarthik Alagappan

KAGGLE COMPETITION: HOUSE PRICES - ADVANCED REGRESSION TECHNIQUES	1
Abstract:	1
Introduction:	1
Important Definitions:	3
Overview Of Proposed Systems:	6
Technical Details of Proposed Approach or System:	8
Experiments:	8
Conclusion:	12

Abstract:

In this paper, we have applied multiple machine learning techniques to forecast the sale price of properties. Numerous elements influence housing sales prices, including material quality, living area square feet, garage size, house location, number of bedrooms, and so on. We concentrate on aspects that are typically considered by applicants. We use data from the Kaggle website. The prediction model for houses is then built using machine learning algorithms such as Random Forest, Decision Regressor, Extra Trees Regressor, advanced regression techniques such as Light GBM, Cat Boosting, XGBoost, Linear Regressor, RANSAC regressor, Support Vector Regressor, Lasso Regressor models. We fine tuned the models with different parameters and compared these methods using the RMSLE and 10-fold cross validation results. In this paper we have reported our experimentation methods and the results we obtained from those experiments.

Keywords:

Machine Learning, Advanced regression, Random forest, CatBoost.

Introduction:

Real estate is not only a vital part of the national economy, but it is also a big source of concern for citizens. People's attention to house prices continues to rise as a result of rising housing demand. It is vital to provide accurate house price forecasts. Multiple factors influence housing prices, including time and space, house age, surrounding conditions, communities, transportation, and so on. Existing prediction models are often single predictors, i.e., the

prediction is made using only one forecasting model. When datasets are noisy, the prediction accuracy of this type of model is poor.

We have gone through two references, for one they have used multiple machine learning algorithms to train the data and for another they have used the same algorithm throughout the project. Both of them used RMSLE for loss function.

We have chosen regression techniques as our baseline models due the following reason:

In predictive analytics, machine learning regression models are primarily used to forecast trends and predict outcomes. To understand the link between different independent variables and a result, regression models will be trained. As a result, the model can comprehend the numerous aspects that may lead to a desired conclusion. The models that result can be used in a variety of locations and in a variety of ways. Market variations can be forecast and accounted for, and campaigns can be tested by adjusting different independent factors.

We looked at both internal and external parameters in this article, such as the number of bedrooms, living area, and construction material (such as location, proximity, upcoming projects, etc.). Then, using these parameter values, we compared different machine learning techniques, including Random Forest, Decision Regressor, Extra Trees Regressor, advanced regression techniques such as Light GBM, Cat Boosting, XGBoost, Linear Regressor, RANSAC regressor, Support Vector Regressor, Lasso Regressor using Root Mean Squared Log Error (RMSLE). We'll also talk about the significance of our approach and the methods we used.

All the experiments were run using open-source software program Python3 and in Jupyter Notebook. Various data analyses were done on the data set, as shown in our experiment. The process of picking a subset of variables from a given set of parameters, either based on their importance or frequency, is known as feature selection. Feature extraction, on the other hand, is the process of lowering the data's dimensionality. The initial set of data is transformed into non-redundant, equally informative derived values. Random Forest, Decision Regressor, Extra Trees Regressor, advanced regression techniques such as Light GBM, Cat Boosting, XGBoost, Linear Regressor, RANSAC regressor, Support Vector Regressor, Lasso Regressor using RMSLE are the feature selection algorithms employed, and the mean of each algorithm is obtained.

We have collected the data from the Kaggle website. We have used train.csv , test.csv, data description.txt and sample submission.csv.

For our projects, we have taken the following steps as mentioned. Components of our machine learning Project:

1. Data collection
2. Data Exploration and Preprocessing:
3. Data Exploration
4. Data Cleaning
5. Label encoding
6. Data Reduction
7. Data Splitting:

8. Splitting data into two parts training and testing datasets
9. Applying the algorithms
10. Training and Comparing different machine learning algorithms with fine tuned classifiers
11. Evaluating the models with test dataset and testing Kaggle's dataset.

Based on our results from testing and comparing different types of regression models, we found that CatBoost has the best performance and placed us in the top 11% of the Kaggle competition at the time of this writing.

452

Ronaldo Engelke Cunha



0.12333

31

1m



Your Best Entry!

Your most recent submission scored 0.12333, which is an improvement of your previous score of 0.12408. Great job!

Tweet this

We finished the competition at position 452 out of 4173 teams.

Important Definitions:

Data: Any unprocessed fact, value, text, sound, or image that has not been understood and examined qualifies. The most critical component of Data Analytics, Machine Learning, and Artificial Intelligence is data. We can't train any model without data, therefore all of today's research and automation will be for naught.

The dataset utilized in this experiment was a Kaggle open source dataset. Houses (i.e. observations) were accompanied with attributes (i.e. features, variables, or predictors) and the sales price for each house in our training data set. The residences in our testing set all had the same characteristics, but the sales price was not included because it was our goal variable. However, only those parameters were picked that are likely to have an impact on house prices. Area in square meters, for example. Overall quality, which assesses the house's overall condition and finish. The location of the house, the year it was built, and the size of the house are all factors to consider. The number of bedrooms and baths, as well as the total number of people who will be staying in the house, The size of the garage and the number of cars that can fit in it, the size of the swimming pool, the house's selling year, and the price at which it was sold are all factors to consider. The selling price is a dependent variable that is influenced by a number of other factors. We'll make predictions using the test.csv data once we've trained a model using the train.csv data. We'll just go over our main findings, which will help us with feature engineering. We have categorical qualities that must be converted to dummy variables.

Prediction Target: The feature of a dataset about which we wish to learn more is known as the target variable of a dataset. A supervised machine learning algorithm learns patterns and uncovers links between other features in the dataset and the target using previous data.

SalePrice: the target variable we are trying to predict is Sale Price

Variables or concepts in data: A feature is a trait of the thing you're analyzing that can be measured. The basic building elements of datasets are features. The quality of the features in your dataset has a big influence on the quality of the insights you'll get from machine learning with it. Furthermore, different business problems within the same industry may not necessitate the same features, which is why it is critical to have a clear grasp of your data science project's business goals.

Table 1 : Feature Descriptions

Feature	Description	Feature	Description
SalePrice	the property's sale price in dollars - The predicting value	Heating	Type of heating
MSSubClass	The building of class	HeatingQC	Heating quality and condition
MSZoning	The general zoning classification	CentralAir	Central air conditioning
LotFrontage	Linear feet of street connected to property	Electrical	Electrical system
LotArea	Lot size in square feet	1stFlrSF	First Floor square feet
Street	Type of road access	2ndFlrSF	Second floor square feet
ExterQual	Exterior material quality	Functional	Home functionality rating
LowQualFinSF	Low quality finished square feet (all floors)	SaleType	Type of sale
LotShape	General shape of property	GrLivArea	Above grade (ground) living area square fee
LandContour	Flatness of the property	BsmtFullBath	Basement full bathrooms
Utilities	Type of utilities available	BsmtHalfBath	Basement half bathrooms
LotConfig	Lot configuration	FullBath	Full bathrooms above grade
LandSlope	Slope of property	HalfBath	Half baths above grade
Neighborhood	Physical locations within Ames city limits	Bedroom	Number of bedrooms above basement level
Condition1	Proximity to main road or railroad	Kitchen	Number of kitchens
Condition2	Proximity to main road or railroad (if a second is present)	KitchenQual	Kitchen quality
BldgType	Type of dwelling	TotRmsAbvGrd	Total rooms above grade (does not include

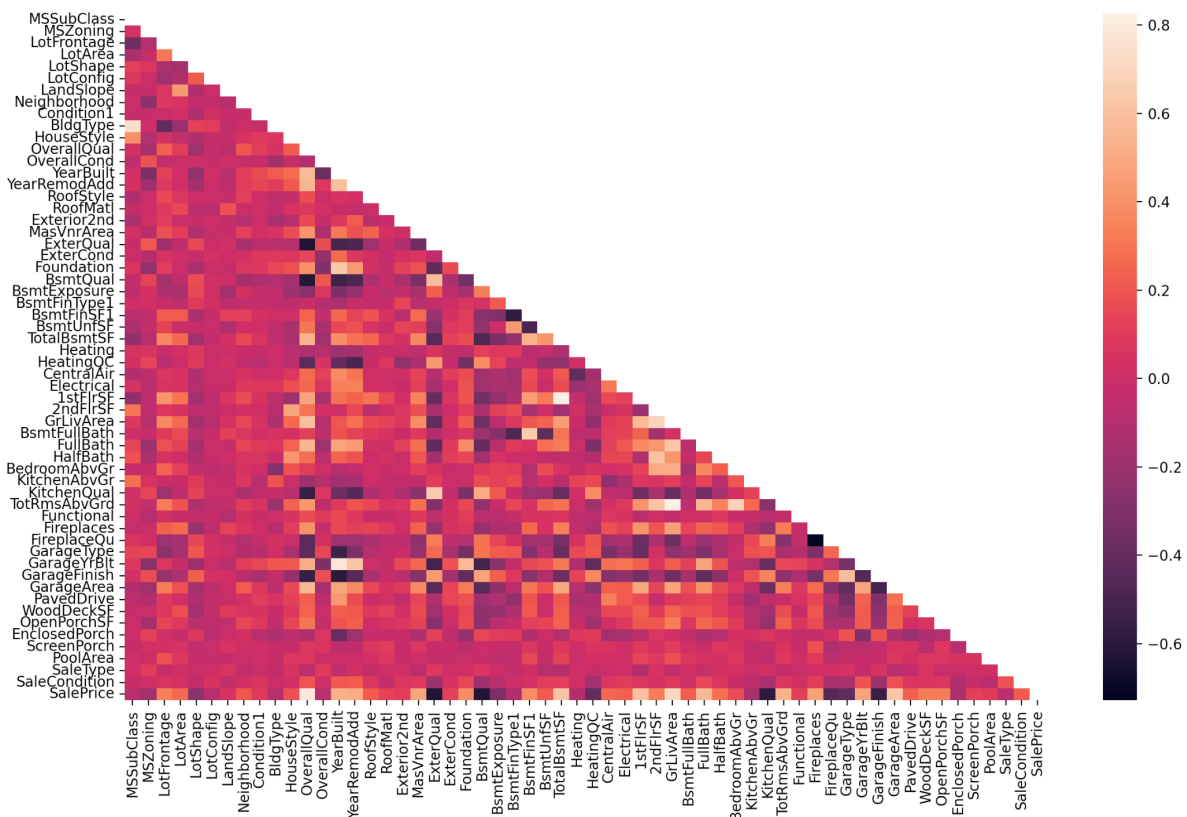
			bathrooms)
HouseStyle	Style of dwelling	Fireplaces	Number of fireplaces
OverallQual	Overall material and finish quality	FireplaceQu	Fireplace quality
OverallCond	Overall condition rating	GarageType	Garage location
YearBuilt	Original construction date	GarageYrBlt	Year garage was built
YearRemodAdd	Remodel date	GarageFinish	Interior finish of the garage
RoofStyle	Type of roof	GarageCars	Size of garage in car capacity
RoofMatl	Roof material	GarageArea	Size of garage in square feet
Exterior1st	Exterior covering on house	GarageQual	Garage quality
Exterior2nd	Exterior covering on house (if more than one material)	GarageCond	Garage condition
MasVnrType	Masonry veneer type	PavedDrive	Paved driveway
MasVnrArea	Masonry veneer area in square feet	WoodDeckSF	Wood deck area in square feet
ExterQual	Exterior material quality	OpenPorchSF	Open porch area in square feet
ExterCond	Present condition of the material on the exterior	EnclosedPorch	Enclosed porch area in square feet
Foundation	Type of foundation	3SsnPorch	Three season porch area in square feet
BsmtQual	Height of the basement	ScreenPorch	Screen porch area in square feet
BsmtCond	General condition of the basement	PoolArea	Pool area in square feet
BsmtExposure	Walkout or garden level basement walls	BsmtFinType1	Quality of basement finished area
BsmtFinSF1	Type 1 finished square feet	SaleCondition	Condition of sale
BsmtFinType2	Quality of second finished area (if present)	MiscVal	\$Value of miscellaneous feature
BsmtUnfSF	Unfinished square feet of basement area	MoSold	Month Sold
BsmtFinSF2	Type 2 finished square feet	YrSold	Year Sold

TotalBsmtSF	Total square feet of basement area		
-------------	------------------------------------	--	--

Problem Statement: People who are trying to buy a new home are more cautious about their budgets and market strategy. House price forecasting can assist customers in determining the best time to buy a home. Everything is moving away from manual to automated processes nowadays. As a result, we decided to create a model that estimates the price of a property based on the user's input of location and size.

Overview Of Proposed Systems:

We have used pairwise correlation to remove highly correlated data(not with the value of y). We are also retaining features with a correlation factor of 0.05 or more with the predicting data. We found boosting-models to fare extremely well with any feature that's at least slightly correlated with sales price - whereas bagging/bootstrapping models, like Random Forest Regressor, fared better with features showing 40~50% correlation with the target variable. Even when hyperparameter tuned, Random Forest Regressor could not reach the same 10-fold average RMSLE scores of XGBoost and CatBoost - thus we shifted our focus to preparing our data, preprocessing, and tuning for boosting models in mind, rather than bagging models.



Pairwise Correlation Matrix

We have undergone model testing and hyperparameter tuning for the following learning models:

- **Linear Regression Algorithms:**
 - RANSAC Regression
 - Linear Regression (default)
 - Logistic Regression
 - Lasso Regression
- **Ensemble Boosting-based Algorithms:**
 - AdaBoost
 - CatBoost
 - XGBoost
 - Light GBM
- **Ensemble Bagging-based Algorithms:**
 - Random Forest Regressor (our case study)
- **Non-Ensemble Tree-based:**
 - Decision Tree Regressor

We explored various families of models in the hopes of establishing a baseline for performance and hyperparameter tuning, while also changing the composition of our training dataset.

Potential Issues:

- Finding optimal inclusion of features in training based on correlation varies from bagging-boosting model type.
- Hyper parameter tuning for various model groups also varies depending on average feature correlation - and inclusion of features with low correlation.
- There exists different levels of optimal inclusion of features in testing the models - meaning performance and parameters for models belonging to the bagging/boosting family are optimized with different density/number of correlated features included.
- This makes hyperparameter tuning a time-expensive task, and one which we must iteratively try with varying levels of included features.

Improved Methods:

- CatBoost has given us the greatest performance, and has given us our best placement thus far in the competition.
- We have seen a sharp decline in avg 10-fold RMSLE in all models after fine-tuning the parameters and tweaking preprocessing methods (since we have started), especially with the boosting models.
- Random Forest and the boosting-type models, have seen considerable drop in 10-fold average RMSLE and considerable increases in overall accuracy.

Technical Details of Proposed Approach or System:

For our project we have followed the steps as described below:

1. Model Discover/Importation:

- a. We had multiple features that played a part in determining the final sale price, most of which were independent of each other. So we decided to stick with a regressor model as our main types of ML models to train and test
 - i. Random forest was our first and baseline model that we trained and tested
 - ii. We also compared how random forest model performed against other regression models
- b. We used the Scikit-learn package to import pre-built models and used the data provided by Kaggle to train and test those models.

2. Data Preprocessing:

- a. First, we have checked for null values and removed features whose null value to actual value percentage was more than 75%.
 - b. Then we have used label encoding to convert categorical variables
 - c. For the remain features with null values, we filled the empty values with the mean of the respective feature
 - d. We have used pairwise correlation to find the correlation between variables. If two variables have a correlation of 85% or higher, then we drop one.
 - e. Then we calculate the correlation of the retained features with the predicting feature and retained features where the correlation was 5% or higher.
 - f. We have then visualized our current dataset.
 - g. That concludes our data preprocessing steps.
3. We split the dataset into two parts- training and testing data.
 4. After tuning the data preprocessing, we went to train the model with different machine learning algorithms and tune that.

We have undergone model testing and hyperparameter tuning for the models mentioned previously. Among them, CatBoost gave us the best result (though XGBoost gave higher accuracy but for CatBoost the RMLSE was lower so we chose that). For that we have used: RMSLE as loss function, 750 iterations, 10-fold average RMSLE, 0.04 learning rate. Using that we got an accuracy rate of 89% with the 10-fold average being 0.123.

Experiments:

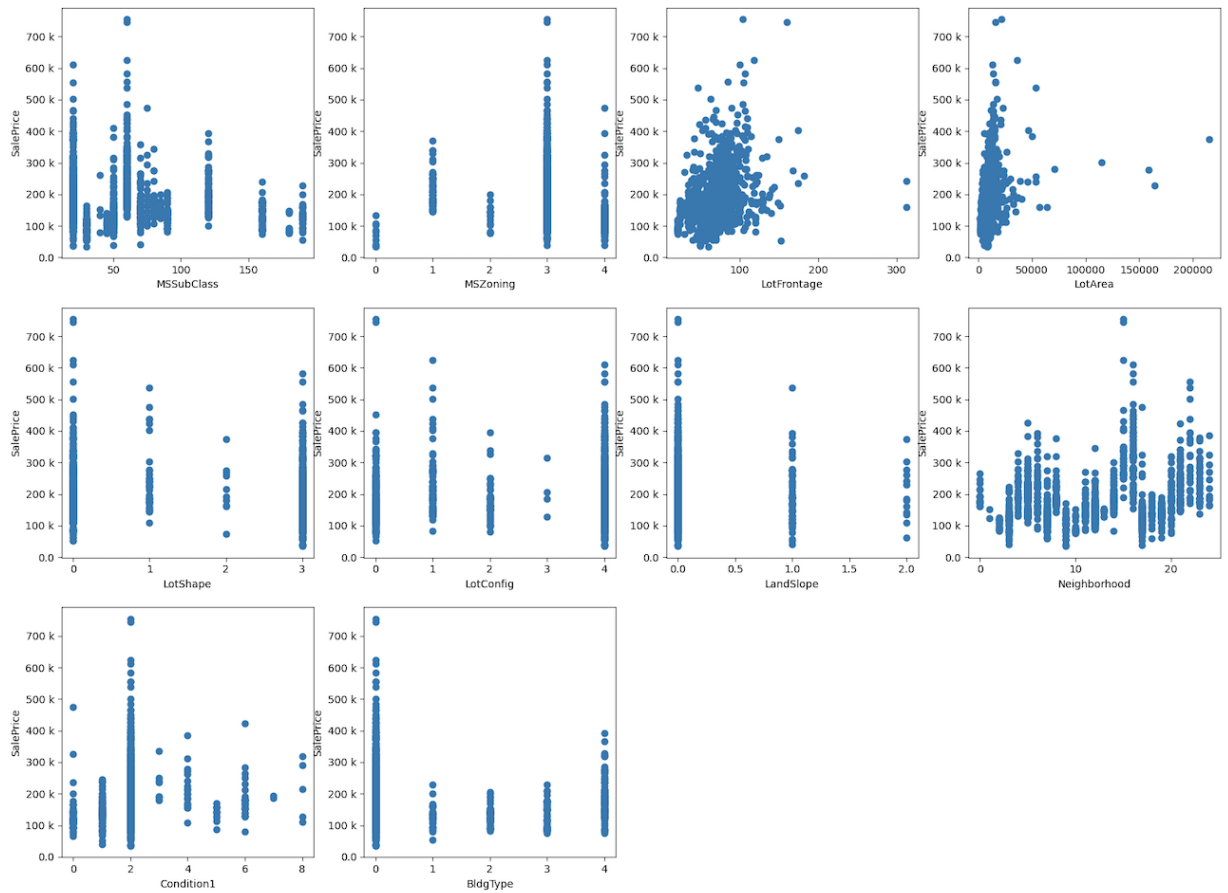
For our initial model, we tested with just RandomForest model as our baseline model to compare with. We designed our preprocessing steps, model creation, and hyperparameter tuning around the requirements of a bagging/bootstrapping model. Our preprocessing steps were simple in the sense we hand chose certain features by checking which features logically correlate to the

final sale price in the real world, discarded a lot of features with null values, and filled missing values with the mean. Then initially we just trained the RandomForest model with that preprocessor data, with which we got 84% accuracy. Then moving forward, we reworked the preprocessing stage such that we used correlation factor to remove and retain features, which showed a good improvement in the scores. We also noticed increased performance from using boosting models over bagging, so we changed our preprocessing approach by lowering the thresholding for feature correlation, from 50% to 5% - a choice which further improved the performance of the boosting models.

For our regression models we have applied hyperparameter tuning: All models (except linear regression and DT models) underwent a Randomized Search CV over a parameter grid of impactful model parameters, with a stratified k-fold approach of 100-150 iterations per fold.

We saw a few noteworthy relationships/conclusions:

- All models generally see lowered 10-fold mean RMSLE performance (higher values) when we retain labels with >50% only.
- All models tend to perform better by including features that share even just 10% correlation with the regression target, while retaining features with feature-y_value correlation higher than 5% give the best results.
- Random Forest tends to perform better than most boosting models in either case (though it is the most tuned of all currently)
- Of all the boosting models, we have seen the best results from CatBoost, XGBoost and Light GBM.
- Parameter tuning varies drastically depending on average feature correlation (impacted by choosing different features to retain based on label correlation) in testing.
- CatBoost is currently the most performant, giving us the lowest RMSLE and 10-fold average RSMLE score.



Scatter plot of first 10 retained features (Y value is SalePrice) - total of 74 features were retained

Here are a few screenshots of results we've attained with varying models. These models, except for the Linear Regression models, have gone through rounds of hyperparameter tuning, though they are not perfect by any means. The goal of this project and this competition is to essentially achieve the lowest average RMSLE score, assuring the lowest levels of aggregate error in regression modeling. The metric with the highest correlation of accuracy here is the five-fold avg RMSLE, as it is the best, most robust predictor of overall model performance. The screenshots demonstrate the scores we got and are in chronological order of fine-tuning (our top 4 models).

```
XGBoost Regressor
10-Fold Avg RMSLE 0.1264556876468248
Current Run Mean RMSE: 29153.07948674257
R2 Coefficient Score: 0.91
Accuracy (100 - MAPE): 92.05 %
RMSLE 0.1134154153583645
```

```
CatBoost Regressor
10-Fold Avg RMSLE 0.12496140985196504
Current Run Mean RMSE: 28713.44798208351
R2 Coefficient Score: 0.91
Accuracy (100 - MAPE): 91.56 %
RMSLE 0.11780232563969793
```

```
Random Forest Regressor
10-Fold Avg RMSLE 0.142109551640126
Current Run Mean RMSE: 30403.81983160282
R2 Coefficient Score: 0.9
Accuracy (100 - MAPE): 90.48 %
RMSLE 0.14080639202885348
```

```
AdaBoost Regressor
10-Fold Avg RMSLE 0.20299875364704498
Current Run Mean RMSE: 38843.088196405944
R2 Coefficient Score: 0.83
Accuracy (100 - MAPE): 84.35 %
RMSLE 0.2094264046813808
```

Since CatBoost showed the best results, we tried to fine-tune the CatBoost model with new preprocessing methods and parameters such as creating new features like total square footage and total bathrooms, but they did not yield improved results. We also tried one-hot encoding as opposed to label encoding done before, but that didn't help with score improvement either as shown below. A possible reason why one-hot encoding did not render better preprocessing results with boosting models is the sheer number of additional columns added to the training data. We also attempted to reduce overall skewness of the predicting variable and that also resulted in a decline in performance for our boosting models, but increased the performance of the linear regression model. This change, along with completing a log-transformation of the dataset, made the data linearized, which greatly improved the performance of all our linear regression models (linear regressor, RANSAC, Lasso). However, none of these models matched the earlier performance of CatBoost with non-linearized data.

```
CatBoost Regressor
10-Fold Avg RMSLE 0.1309307106825448
Current Run Mean RMSE: 26672.6007211824
R2 Coefficient Score: 0.92
Accuracy (100 - MAPE): 91.9 %
RMSLE 0.10840876408840437

Linear Regressor
10-Fold Avg RMSLE 0.15816430563430975
Current Run Mean RMSE: 31761.689540388405
R2 Coefficient Score: 0.89
Accuracy (100 - MAPE): 89.3 %
RMSLE 0.14404232839036843
```

We designed our pipeline, especially the preprocessing stage, with the goal of achieving the best possible performance from a single boosting model, and not for linear regression, or stacking models. Had we attempted to preprocess the data with the goal of optimizing linearized models or multiple ensemble models in mind, we could have potentially seen different overall results in the end.

Related Work:

- [A Detailed Regression Guide with House-pricing](#)

In this project they have done: through write-ups, learning, reviewing, and explaining

complex data science concepts, performed a thorough data analysis as well as visualizations.

Create models : XGBoost, Random Forest, Decision Tree, Ridge Regressor, and Lasso Regressor models that are capable of predicting higher house sale prices. Also they have used RMSLE. Lastly they blended all the models to predict the RMSLE score on training data.

So, it was a great overview of different models to predict the house prices.

- [Getting Started with Kaggle: House Prices Competition](#)

In this project they have acquired the data, explored it, performed feature engineering and transformed the features and the target variable. Then they built a model, predicted and submitted the predictions.

Here, they have used RMSLE, built a linear model and finally used Ridge regularization to remove less important features of the data. It was also a good review on predicting house models though they have not considered multiple training algorithms.

Conclusion:

We employed machine learning techniques to estimate housing values in this project. The process for analyzing the dataset and determining the correlation between the parameters has been described in detail. As a result, we can choose parameters that are not associated with one another and are independent in nature. These feature sets were then fed into multiple algorithms, resulting in a csv file with estimated house prices. As a result, we compared the performance of each model using the RMSLE metric. This project work can be valuable for the development of applications for diverse cities by improving the error values.

We created the final model by testing multiple preprocessing methods and fine tuning multiple regression models. Testing multiple types of models and continually training them proved to be the best method to find an optimal model. Our aim was for the models to show improvement with each iteration, resulting in a smaller 10-fold average RMSLE score. Based on the presented experiments and results, CatBoost proved to be the best one to predict the final sale price of the houses and placed us in the top 11% of the Kaggle competition.

You can find our code here:

https://colab.research.google.com/drive/1AFibFv_ky6l6Gs-2wAYV6cV1HDHx5H05?usp=sharing

References:

[https://www.kaggle.com/code/masumrumi/a-detailed-regression-guide-with-house-pricing-notebook#Fitting-model\(simple-approach\)](https://www.kaggle.com/code/masumrumi/a-detailed-regression-guide-with-house-pricing-notebook#Fitting-model(simple-approach))

<https://www.dataquest.io/blog/kaggle-getting-started/>

<https://towardsdatascience.com/how-to-use-pairwise-correlation-for-robust-feature-selection-20a60ef7d10>

<https://towardsdatascience.com/catboost-regression-in-6-minutes-3487f3e5b329>

<https://medium.com/@sergei740/hyperparameter-tuning-lightgbm-using-random-grid-search-dc11c2f8c805>